

Development of an Automatic Design Optimization Platform

for a 2D Heat-Conduction System Using FreeFEM++

HEAT-COND benchmark — FreeFEM++ and Python

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Abstract

We develop a compact, automatic design-optimization platform for the 2D steady-state HEAT-COND heat-conduction benchmark, coupling a FreeFEM++ P_1 finite-element solver with a Python optimization driver. Through seven systematic studies we show that the objective is smooth and unimodal with a *boundary* optimum at $\mathbf{x}^* = (1, 1, 1, 1, 1, 0.01)$, giving $J^* \approx 0.730$ at mesh density 50 — roughly an 850% improvement over the uniform design. Of the three compared methods (Nelder–Mead, Differential Evolution, and an Adam $\rightarrow$ L-BFGS hybrid), **Nelder–Mead is the most efficient and is chosen as the platform default**; Differential Evolution serves as a global-optimality check, and the gradient hybrid illustrates the value of adaptive preconditioning on this ill-conditioned problem. We further extend the benchmark with a material-cost objective whose Pareto front shows that 32% of the conductive material can be saved for under 1% performance loss.

1 Introduction and Project Objective

This project develops a compact, automatic design-optimization platform for a two-dimensional steady-state heat-conduction problem, the HEAT-COND benchmark drawn from the reduced-basis literature [1]. The platform integrates, within a single reproducible workflow, the five required components: a finite-element solver written in FreeFEM++, an objective-evaluation module, a Python optimization driver, a graphical user interface, and a numerical-analysis framework for comparing optimization strategies.

The scientific goal is to couple the solution of an elliptic partial differential equation (PDE) posed on a non-trivial 2D geometry with an automated parametric optimization loop, and to determine, through systematic experiments, *which optimization strategy is best suited to this problem and why*. As we show, the objective functional is smooth and unimodal with its optimum located on the boundary of the admissible set; this single structural fact explains the entire ranking of the methods we tested, and motivates the optimizer chosen as the platform default.

2 Mathematical Model

2.1 Geometry

The computational domain $\Omega \subset [0, 1]^2$ is a comb-shaped region consisting of a central vertical *spine* of width $w_s = 0.10$ and five pairs of horizontal *fins* of thickness $t_f = 0.06$, centred at $y \in \{0.16, 0.32, 0.48, 0.64, 0.80\}$. The inner spine edges are at $x_g = (1 - w_s)/2 = 0.45$ and $x_d = x_g + w_s = 0.55$. Two boundary regions are distinguished: Γ_{Base} , the bottom edge of the spine $\{(x, 0) : x \in [x_g, x_d]\}$, and Γ_{Fin} , the remainder of $\partial\Omega$ (the spine top, the outer fin tips, and all fin flanks). The domain, its boundary labels and the P_1 mesh are shown in Fig. 1.

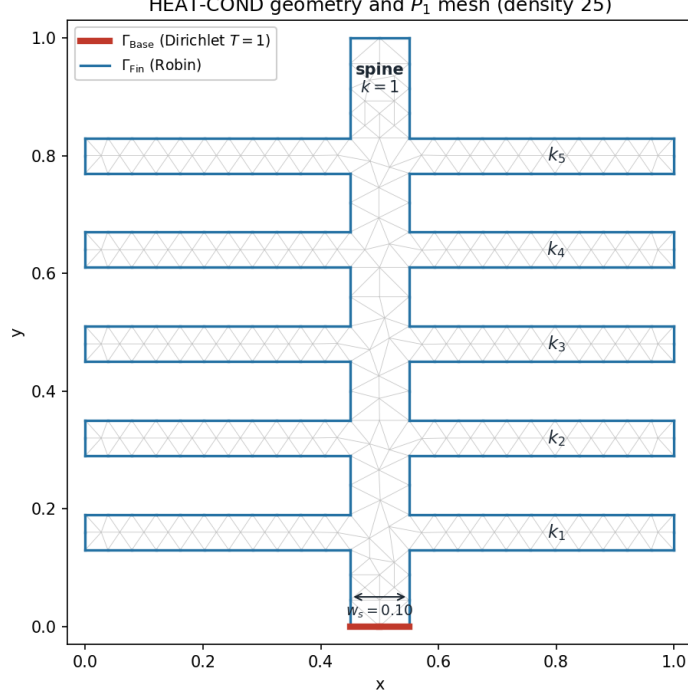


Figure 1: HEAT-COND geometry and P_1 mesh (density 25). The red base Γ_{Base} carries the Dirichlet condition $T = 1$; the blue boundary Γ_{Fin} carries the Robin condition. The central spine has conductivity $k = 1$; each fin i has conductivity k_i .

2.2 Governing Equation

The dimensionless temperature T satisfies the steady-state heat-conduction equation with mixed (Dirichlet–Robin) boundary conditions:

$$\begin{cases} -\nabla \cdot (k(\mathbf{x}) \nabla T) = 0 & \text{in } \Omega, \\ T = 1 & \text{on } \Gamma_{\text{Base}}, \\ k(\mathbf{x}) \frac{\partial T}{\partial n} + \text{Bi} T = 0 & \text{on } \Gamma_{\text{Fin}}. \end{cases} \quad (1)$$

The conductivity $k(\mathbf{x})$ is piecewise constant: $k = 1$ in the spine, and $k = k_i$ ($i = 1, \dots, 5$) in fin i , counted bottom to top. The Biot number Bi controls the convective coupling on Γ_{Fin} .

2.3 Objective and Optimization Problem

The objective is to maximize the average temperature on the fin boundary,

$$J(\mathbf{x}) = \frac{1}{|\Gamma_{\text{Fin}}|} \int_{\Gamma_{\text{Fin}}} T \, d\Gamma, \quad (2)$$

over $\mathbf{x} = (k_1, \dots, k_5, \text{Bi}) \in \mathbb{R}^6$ with $k_i \in [0.1, 1.0]$ and $\text{Bi} \in [0.01, 1.0]$: $\max_{\mathbf{x}} J(\mathbf{x})$ subject to box constraints. Physically, a high J means the spine transfers the heat injected at the base ($T = 1$) efficiently to the peripheral fin surfaces.

3 Numerical Formulation and FreeFEM++ Implementation

3.1 Mesh Generation

The mesh is constructed explicitly with FreeFEM’s `buildmesh` rather than by truncating a background grid, so that element edges align exactly with the geometry. The boundary is described as a

closed, counter-clockwise polyline of 44 segments / 45 points, traversed by a single indexed `border`; the number of subdivisions of each segment is proportional to its Euclidean length times a global density `meshsize`. Two boundary labels are assigned: **label 1** for the base segment (Dirichlet) and **label 2** for the rest of the boundary (Robin). Each mesh is cached as `cache/mesh_<size>.msh`. At density 50 the mesh contains 1151 vertices and 1760 triangles (419 vertices at density 25).

3.2 PDE Solver

`solver.edp` reads its parameters through `getARGV`. The weak form is assembled in the P_1 space V_h : find $T \in V_h$ with $T = 1$ on Γ_{Base} such that, for all test functions v ,

$$a(T, v) = \int_{\Omega} k \nabla T \cdot \nabla v \, d\Omega + \int_{\Gamma_{\text{Fin}}} \text{Bi} T v \, d\Gamma = 0. \quad (3)$$

The Dirichlet condition is imposed strongly via `on(1, T = 1)`; the Robin condition enters naturally as the boundary integral over label 2. The objective $J = \int_{\Gamma_2} T \, d\Gamma / \int_{\Gamma_2} 1 \, d\Gamma$ is computed inside the solver and printed to stdout as `Objective_J=<value>`, removing intermediate text files.

3.3 Python Coupling and Verification

The Python layer drives FreeFEM through `subprocess`: `ensure_mesh` caches meshes and `run_solver` parses the objective from stdout. To verify the implementation, the P_1 weak form was re-implemented in pure NumPy and compared against FreeFEM on identical cached meshes: at density 50 the two solvers agree to six significant digits at the optimal corner — $J(1, 1, 1, 1, 1, 0.01) = 0.730396$ in both — and the vertex/triangle counts match exactly. This cross-check validates the operator and the objective, and lets the optimizer comparison below be reproduced without FreeFEM.

4 Optimization Strategy

The driver wraps every algorithm behind a uniform interface returning a normalized dictionary (`best_J`, `best_x`, `n_eval`, `time`, full `history`); random methods use a fixed seed. After a preliminary screening we retained **three complementary methods**, one per family:

- **Nelder–Mead (NM)** — local, deterministic simplex in dimension 6; gradient-free, sensitive to the start, very cheap.
- **Differential Evolution (DE)** — global, population-based, gradient-free; robust but the most expensive.
- **Adam → L-BFGS-B** — a two-phase gradient hybrid: Adam (finite-difference gradient, per-coordinate adaptive step) drives the iterate into a good region, then L-BFGS-B refines with native bound handling.

Because the objective is a black-box PDE solve, the gradient phase uses central finite differences, costing $2 \times 6 = 12$ extra evaluations per gradient in 6D. The experimental design is implemented in the notebook `comparison_methods.ipynb` as seven self-contained studies; the DE hyperparameter study averages each configuration over three independent seeds.

5 Numerical Results

The reference initial design is $\mathbf{x}_0 = (0.5, \dots, 0.5)$, for which $J_0 \approx 0.077$. Unless stated otherwise, comparisons use mesh density 25; the ranking is identical at density 50.

Table 1: Comparison of the three methods from the common start $\mathbf{x}_0 = 0.5$ (density 25).

Method	J^*	n_{eval}	Time (s)	Design $\mathbf{x}^*$
Nelder–Mead	0.73417	115	0.28	(1, 1, 1, 1, 1, 0.01)
Adam $\rightarrow$ L-BFGS	0.73417	373	0.90	(1, 1, 1, 1, 1, 0.01)
Differential Evolution	0.73411	768	1.86	(1, 0.996, 0.986, 1, 1, 0.01)

5.1 Comparison of the Three Methods and the Optimal Design (Study 1)

All three methods reach the same global optimum $\mathbf{x}^* = (1, 1, 1, 1, 1, 0.01)$ — all conductivities at their upper bound and the Biot number at its lower bound, a vertex of the admissible box. At density 50 this corner gives $J^* = 0.730396$, an $\approx 850\%$ improvement over J_0 . Nelder–Mead reaches it at the lowest cost; Differential Evolution is the most expensive and stops just short of the exact corner (Fig. 2). The effect of the optimization on the temperature field is shown in Fig. 3: the initially cold fins (T down to 0.003 at the tips) become uniformly hot ($T \geq 0.633$).

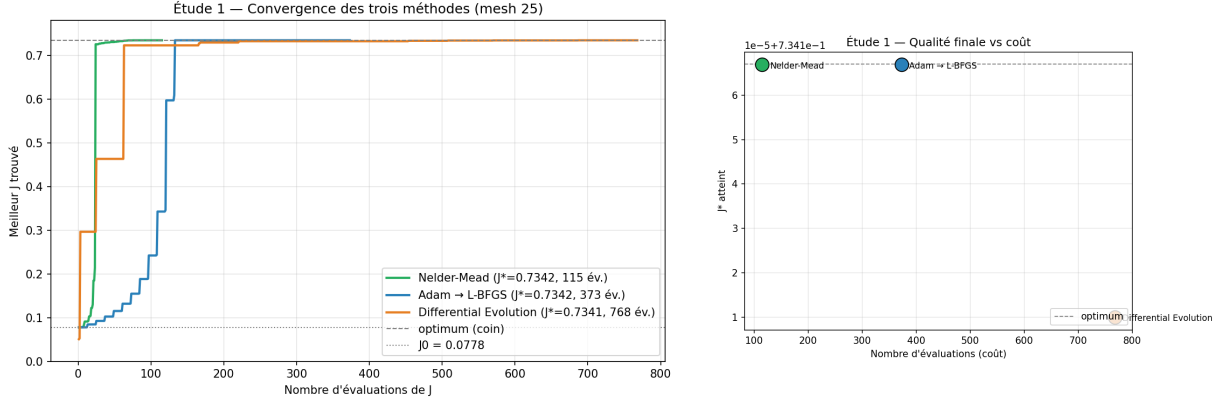


Figure 2: Study 1 — convergence (left) and quality-vs-cost trade-off (right). All three methods reach the optimal corner; Nelder–Mead does so in the fewest evaluations.

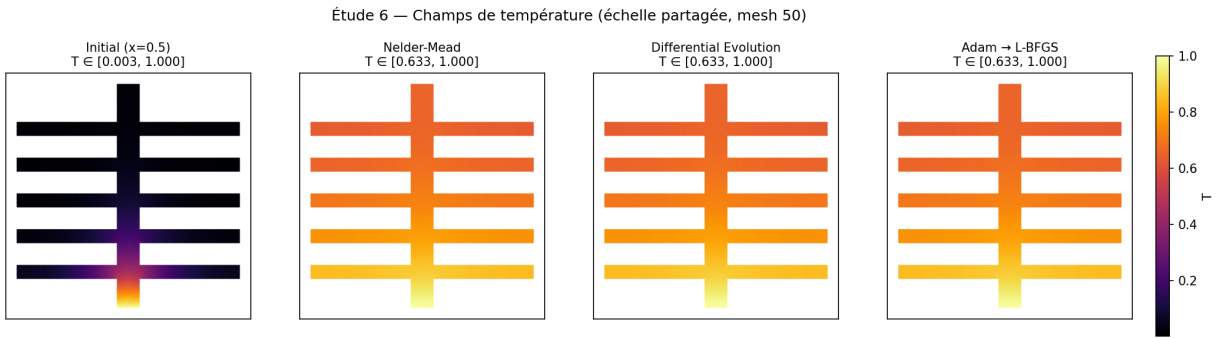


Figure 3: Temperature fields (shared colour scale, mesh 50) for the initial design and the three optimized designs. Optimization raises the minimum fin temperature from 0.003 to 0.633.

5.2 Mesh-Sensitivity (Study 2)

J^* decreases monotonically and stabilizes near 0.73 as the mesh is refined, indicating approximate mesh-independence by density 50–60; coarse meshes slightly over-estimate the objective. The wall-clock time grows steeply with density (more degrees of freedom), as shown in Fig. 4.

Table 2: Optimized objective (Nelder–Mead) vs mesh density.

Mesh density	15	25	40	60
J^*	0.7414	0.7342	0.7313	0.7295

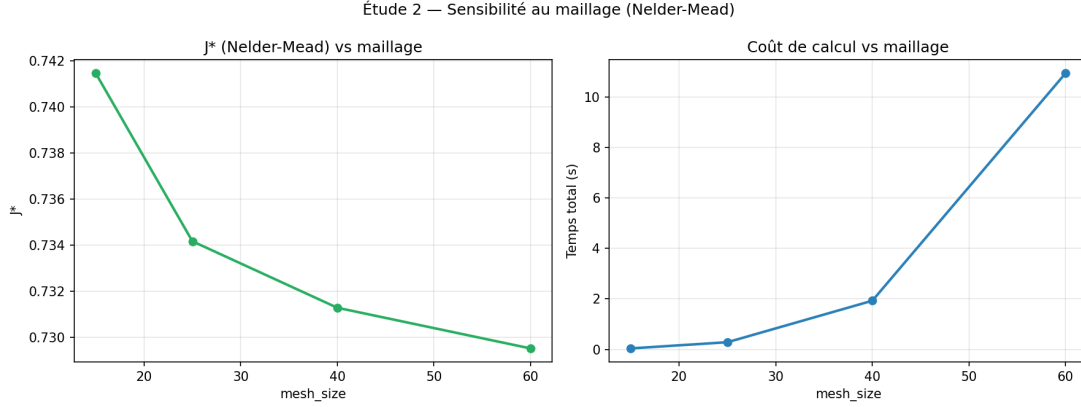


Figure 4: Study 2 — mesh-sensitivity of J^* (left) and computational cost (right).

5.3 Robustness (Studies 3 and 4)

Study 3 varies the starting point for the local/gradient methods (NM, Adam); Study 4 varies the random seed for the only stochastic method (DE). Nelder–Mead is the most reproducible across starting points (standard deviation $\sim 10^{-4}$, mean 0.7341), confirming that the functional is effectively unimodal. Adam→L-BFGS reaches the corner from every tested start (std $\sim 6 \times 10^{-4}$), and DE shows a small seed-to-seed spread (std $\sim 2 \times 10^{-3}$). All three are robust; Nelder–Mead is the most stable (Fig. 5).

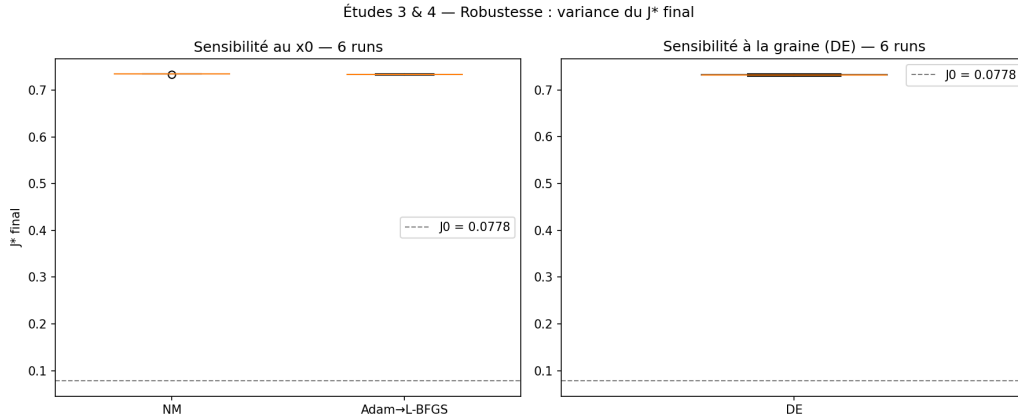


Figure 5: Studies 3–4 — variability of J^* across starting points (NM, Adam) and across seeds (DE).

5.4 Differential-Evolution Hyperparameters (Study 5)

Averaged over three seeds: a larger `popsize` improves both the mean objective and its stability (from 0.672 ± 0.063 at `popsize`= 2 to 0.717 ± 0.002 at 15); a low mutation factor is clearly preferable (0.718 at $F = 0.3$ vs 0.611 at $F = 1.8$); and `best1bin` is the best strategy (0.701). These trends are consistent with an exploitation-dominated, unimodal landscape (Fig. 6).

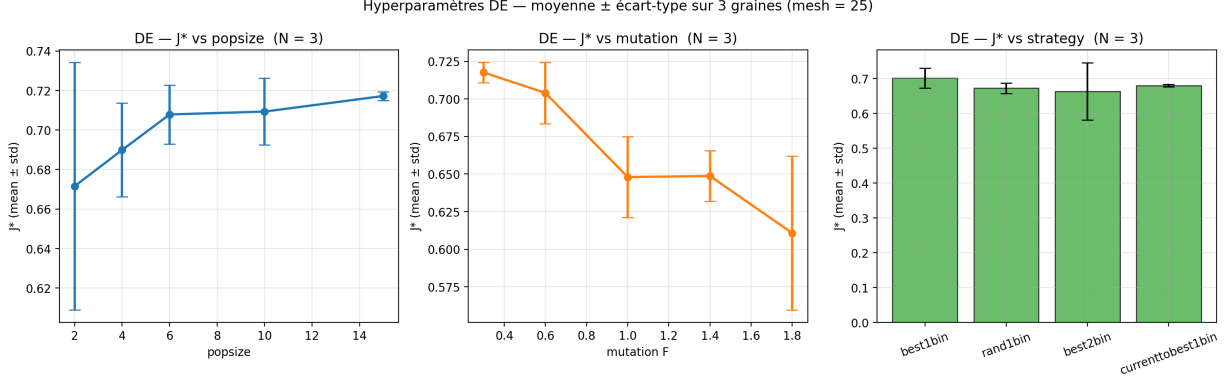


Figure 6: Study 5 — DE hyperparameters (mean $\pm$ std over three seeds): `popsize`, `mutation`, and `strategy`.

5.5 The Adam $\rightarrow$ L-BFGS Hybrid and Preconditioning (Study 7)

This study isolates the contribution of each phase by adding two diagnostic runs from $\mathbf{x}_0 = 0.5$:

Table 3: Diagnostic runs of the gradient hybrid (density 25).

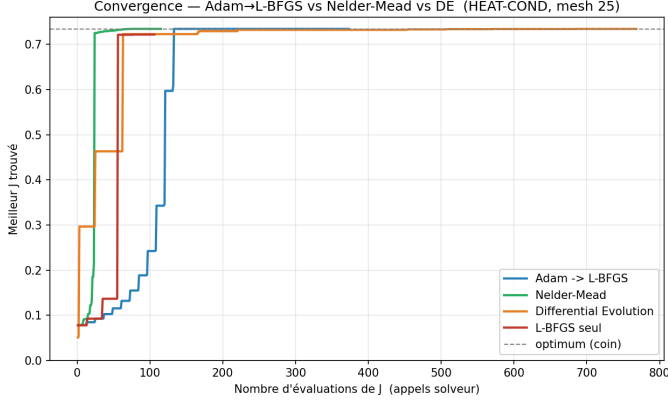
Variant	J^*	n_{eval}	Reaches the corner
Adam $\rightarrow$ L-BFGS	0.734167	373	yes
<i>Adam alone</i>	0.734167	360	yes
<i>L-BFGS-B alone</i>	0.721633	106	no — stalls

L-BFGS-B alone stalls at $J = 0.722$ with the conductivities stuck near 0.5, whereas the hybrid reaches the corner (Fig. 7a). A direct probe of the gradient at the stall point explains why: there $|\partial J / \partial \text{Bi}| \approx 19.4$ but $|\partial J / \partial k_i| \approx 0.009$, a ratio of about 2000:1. Once Bi saturates at its lower bound, the objective is almost flat in the conductivity directions, and the pure quasi-Newton method declares convergence. Adam, which normalizes each coordinate by the running root-mean-square of its gradient, turns this tiny but consistent signal into full-sized steps and drives the conductivities to 1.0. This is a textbook illustration of adaptive preconditioning on an ill-conditioned problem, and it is exactly why the first phase of the hybrid is necessary. The three methods converge to the same physical design (Fig. 7b).

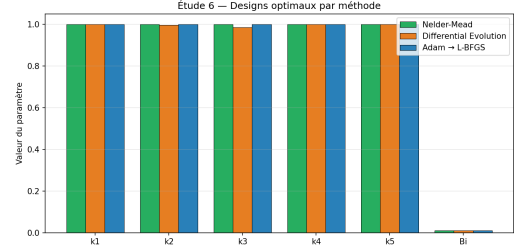
5.6 Extension: Material-Cost Trade-off

The optimum above is a vertex of the box because higher conductivity is free. A more realistic design problem penalizes material: we maximize the scalarized objective $F_\lambda(\mathbf{x}) = J(\mathbf{x}) - \lambda \bar{k}$, where $\bar{k} = \frac{1}{5} \sum_i k_i$ is the mean conductivity (a proxy for material cost), and sweep $\lambda \geq 0$ to trace the performance–cost Pareto front (Fig. 8a). The penalty is evaluated in Python around the solver, so it ports unchanged to FreeFEM and is exposed in the graphical interface through a single λ field.

Two findings stand out. First, the front is strongly concave near the optimum: one can remove 32% of the conductive material for only a 0.8% loss of performance (the marked knee, $\bar{k} = 0.68$, $J = 0.728$ versus $J_{\text{max}} = 0.734$). Second, the optimal design becomes non-uniform with a clear physical ordering $k_1 > k_2 > k_3 > k_4 > k_5$ (Fig. 8b): the fins nearest the hot base retain their conductivity as λ grows, while the upper fins are economized first; the Biot number stays at its lower bound throughout (it carries no material cost). This trade-off turns the otherwise trivial corner problem into a genuine multi-objective design study.

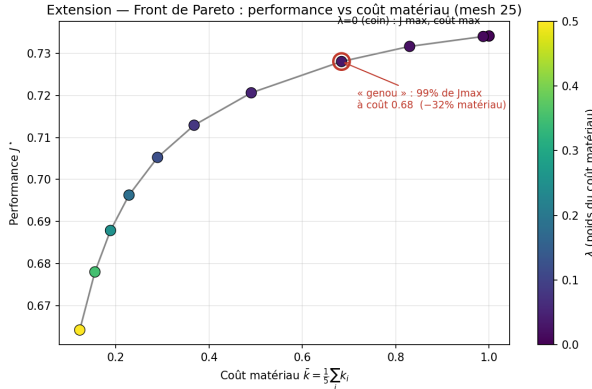


(a)

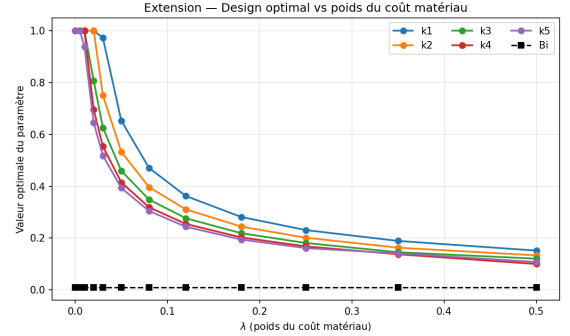


(b)

Figure 7: (a) Study 7 — convergence of the hybrid against NM and DE; L-BFGS alone (dashed) stalls below the optimum. (b) Optimal designs found by the three methods: all converge to $k_i \rightarrow 1$, $Bi \rightarrow 0.01$.



(a)



(b)

Figure 8: Material-cost extension. (a) Performance–cost Pareto front; the marked knee retains 99% of $J_{\max}$ at -32% material. (b) Optimal conductivities versus the cost weight λ : the bottom fins (k_1) stay conductive longest, the Biot number stays at 0.01.

5.7 Optimizer Comparison on the Penalized Problem

Section 5.6 traced the front with a single optimizer. We now repeat the λ sweep with all three methods to check whether the non-trivial (interior) optimum changes the ranking established on the pure-performance problem. Figure 9a overlays the three Pareto fronts: Nelder–Mead and Adam $\rightarrow$ L-BFGS coincide, while Differential Evolution lies marginally inside the front in the mid-cost region, reflecting its fixed, budget-limited search. At a representative weight $\lambda = 0.05$ (near the knee), Nelder–Mead and Adam reach the same scalarized optimum $F^* = 0.696$, with Nelder–Mead the cheapest (178 versus 304 evaluations), while DE attains a slightly lower $F^* = 0.695$ at 324 evaluations (Table 4, Fig. 9b). The conclusion of the pure-performance study therefore carries over to the constrained problem: all three methods agree on the trade-off, and Nelder–Mead remains the most efficient default.

Table 4: Three optimizers on the penalized problem at $\lambda = 0.05$ (mesh 25).

Method	J	cost $\bar{k}$	$F = J - \lambda \bar{k}$	n_{eval}
Nelder–Mead	0.7206	0.490	0.6961	178
Adam $\rightarrow$ L-BFGS	0.7212	0.502	0.6961	304
Differential Evolution	0.7237	0.581	0.6947	324

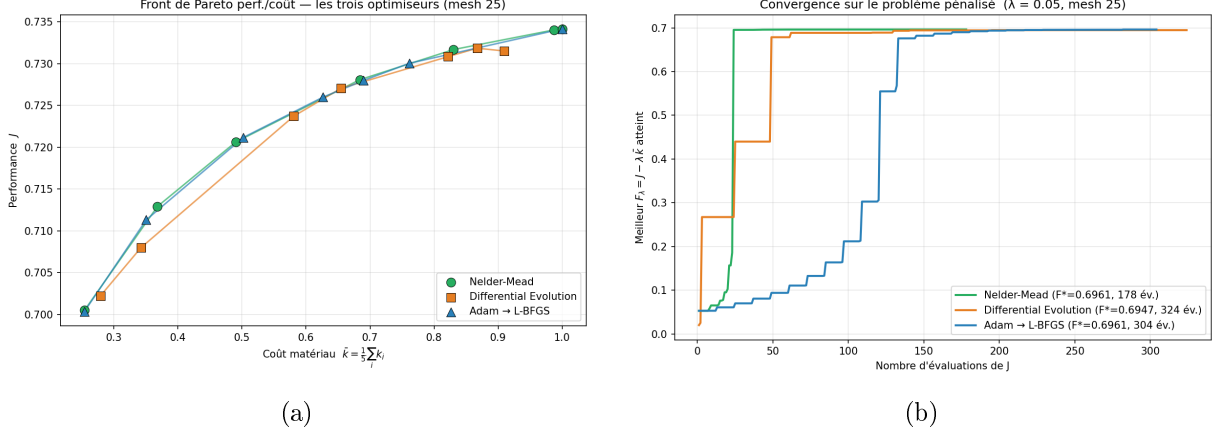


Figure 9: The three optimizers on the material-cost problem. (a) Pareto fronts: Nelder–Mead and Adam $\rightarrow$ L-BFGS coincide; DE lies slightly inside. (b) Convergence of the scalarized objective F_λ at $\lambda = 0.05$: all reach $F^* \approx 0.696$, Nelder–Mead fastest.

6 Analysis and Discussion

Physical interpretation. The optimal design ($k = 1$, $\text{Bi} = 0.01$) is intuitive: maximizing every conductivity lets the spine conduct the base heat to the fins with minimal internal resistance, while minimizing the Biot number minimizes convective loss, so the fins stay hot. The objective is monotone increasing in each k_i and decreasing in Bi ; hence the maximizer sits at a corner of the admissible hypercube, and the functional is smooth and unimodal. This single property explains every result above.

Why Nelder–Mead wins here. For a smooth, unimodal objective with a boundary optimum, a local simplex started from the centre slides directly to the corner. It is the most accurate, the cheapest, and the most reproducible; it is therefore the platform’s recommended default.

Why all three are kept. Nelder–Mead is a *local* method and cannot certify global optimality on its own. Differential Evolution, a global method, lands independently at the same corner; this agreement justifies trusting the result as the global optimum. The Adam $\rightarrow$ L-BFGS hybrid contributes the complementary lesson that adaptive preconditioning is what makes a gradient method succeed here. We emphasize that the Nelder–Mead advantage is *problem-specific*: on a multimodal objective the global method would be preferred; on an ill-conditioned problem without a boundary optimum, the gradient hybrid would be.

Cost and bottlenecks. The dominant cost is the FreeFEM PDE solve invoked through a subprocess ($\approx 0.23\text{s}$ per evaluation at density 50). The finite-difference gradient adds a factor of twelve per gradient in 6D, which is why the hybrid, although effective, is more expensive than Nelder–Mead. The most promising accelerations are an analytical/adjoint gradient and a surrogate model trained on the evaluation history.

7 Conclusion and Possible Improvements

We built an automatic design-optimization platform for HEAT-COND satisfying all required tasks: a geometry-aligned FreeFEM++ P_1 solver (Task 1), an objective module computed in the solver (Task 2), an optimization driver coupling the three retained methods (Task 3), full platform integration with a graphical interface and a command-line pipeline (Task 4), and a structured analysis across seven studies (Task 5). The encouraged extensions — comparison of several optimizers, mesh-sensitivity, parameter-sensitivity, and a gradient hybrid — are all included.

The central finding is that the HEAT-COND objective is smooth and unimodal with a boundary optimum at $\mathbf{x}^* = (1, 1, 1, 1, 1, 0.01)$, giving $J^* \approx 0.730$ at density 50. On this landscape **Nelder–Mead is the most efficient optimizer and the recommended default**, with Differential Evolution retained as a global-optimality check and the Adam→L-BFGS hybrid demonstrating the value of adaptive preconditioning where a pure quasi-Newton method stalls. Beyond the required scope, the material-cost extension of Section 5.6 already turns the benchmark into a genuine multi-objective study with an exploitable Pareto front.

Possible improvements include: (i) a *discrete material-selection* formulation replacing the continuous k_i by a catalogue such as $\{0.1, 0.3, 0.5, 0.7, 1.0\}$; (ii) a *surrogate / response-surface model* (e.g. Kriging) trained on the evaluation history; and (iii) an *adjoint gradient*, which would remove the finite-difference overhead and turn the gradient hybrid into the cheapest option.

Member Contributions

(To be confirmed by the team before submission.) Both members contributed substantially to the methodology. Laurent Zhu focused on the FreeFEM++ solver and meshing, the Python–FreeFEM coupling, and the optimization driver and its studies. Alexandre Le Droucpeet focused on the mathematical model, the graphical interface and visualization, the analysis of the results, and the report.

References

- [1] H. Ballout, Y. Maday, C. Prud’homme. *Nonlinear compressive reduced basis approximation for multi-parameter elliptic problem*. In *Multiscale, Nonlinear and Adaptive Approximation II*, pp. 55–73. Springer, 2024.